AI Code Review Audit Report

Project: **javatest1** • Languages: Java

GROUNDED HEALTH SCORE

75 / 100

SUMMARY

The code contains two critical runtime bugs: a guaranteed division by zero error in `calculateAverage` when called with an empty array, and a potential `ArrayIndexOutOfBoundsException` in `printElement` due to unvalidated user input. Additionally, the `Scanner` resource is not closed, leading to a resource leak.

Verdict: Analysis complete. Review findings above for recommended corrections.

Detailed Findings (1 issues)

Division by Zero in calculateAverage

BUG • CRITICAL

Location: `javatest1.java` (Line 12)

In `main`, the method is called with `new int[]{}` (an empty array). The length of this array is 0. Dividing an integer by 0 in Java throws an `ArithmeticException`.

> **Fix:** Modify `calculateAverage` to return 0 if the array is empty to prevent the crash.

Before:

```
static int calculateAverage(int[] numbers) {
    int sum = 0;
    for (int n : numbers) {
        sum += n;
    }
    return sum / numbers.length;
}
```

After:

```
static int calculateAverage(int[] numbers) {
    if (numbers == null || numbers.length == 0) {
        return 0;
    }
    int sum = 0;
    for (int n : numbers) {
        sum += n;
    }
    return sum / numbers.length;
}
```